Department of Civil Engineering, IIT Madras
CV2090 (Probability & Statistics for Civil Engineers)
Jul–Nov 2026

22 Oct, 2026 | Tutorial-8 | 10:00–11:00 am

Instructions:

1. Make assumptions wherever necessary, and *state* them explicitly.
2. Tutorials are closed-phone, closed-tablet, and closed-LLM.
3. You can bring books and notes, you are welcome to discuss with friends and TAs. However, your submission should be your own work.
4. Submit handwritten solutions on A4 sheets, prepared during the tutorial hours.
5. Submissions will not be accepted after the tutorial hours.

Linear Regression and Correlation Analysis

Q1. **Least-Squares Estimation of Regression Parameters** A civil engineering laboratory measures the tensile force applied to a structural steel specimen (X, in kN) and its corresponding elongation (Y, in mm). The summary statistics obtained from $n = 6$ test points are:

$$\sum x_i = 21.0, \quad \sum y_i = 213.0, \quad \sum x_i^2 = 91.0, \quad \sum y_i^2 = 9819.0, \quad \sum x_i y_i = 947.0$$

(a) Derive/state the Ordinary Least Squares (OLS) expressions for the slope parameter $\hat{\beta}$ and intercept parameter $\hat{\alpha}$.

(b) Compute the estimated regression equation $\hat{y} = \hat{\alpha} + \hat{\beta}x$ predicting elongation based on applied tensile force.

(c) Estimate the expected elongation $\hat{y}$ when an external force of $X = 3.5$ kN is applied. ■

Q2. **Variance Decomposition and Conditional Standard Deviation** In a field study on highway traffic engineering, the daily traffic volume (Y, in tens of thousands) and peak-hour traffic volume (X, in thousands) were recorded over $n = 7$ days. Analysis yielded:

$$\sum (y_i - \bar{y})^2 = 23.44, \quad \sum (x_i - \bar{x})^2 = 32.50, \quad \text{and} \quad \hat{\beta} = 0.82$$

(a) Compute the total variation, explained variation, and unexplained (residual) variation Δ^2 in Y.

(b) Calculate the conditional variance $s^2_{Y|X}$ and conditional standard deviation $s_{Y|X}$. Explain why $s^2_{Y|X}$ loses 2 degrees of freedom $(n - 2)$.

(c) Compare the unconditional sample variance s^2_Y with the conditional variance $s^2_{Y|X}$, and explain why $s^2_{Y|X} < s^2_Y$. ■

Q3. **Confidence Interval for the Mean Response** A geotechnical site investigation evaluates soil consolidation settlement (Y, in mm) against effective overburden stress (X, in kPa). For $n = 25$ samples, the fitted linear regression model is $\hat{y} = -0.14 + 0.435x$, with $s_{Y|X} = 7.78$ mm, $\bar{x} = 22.53$ kPa, and $\sum (x_i - \bar{x})^2 = 41,185.4$ kPa^2.

(a) Write out the analytical formula for the $100(1 - \alpha)\%$ confidence interval of the true conditional mean $\mu_{Y|x_i}$ at a specific location x_i.

(b) Compute the 95% confidence interval for the mean response $\mu_{Y|x}$ at $x_i = 22.53$ kPa (at the sample mean $\bar{x}$).

(c) Compute the 95% confidence interval for the mean response $\mu_{Y|x}$ at $x_i = 150.0$ kPa. Explain geometrically why the confidence interval widens as x_i moves away from $\bar{x}$. ■

Hint: Recall that:

$$T = \frac{\bar{y}_i - \mu_{Y|x_i}}{s_{Y|X}\sqrt{\frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum(x_i - \bar{x})^2}}} \sim t_{n-2}$$

Table Values: $t_{0.025,23} = 2.069$, $t_{0.05,23} = 1.714$.

Q4. Correlation Coefficient and Bivariate Normal Linear Regression In a hydrology monitoring station, precipitation X and runoff Y are modeled as jointly normal random variables with sample statistics computed over $n = 25$ rainstorms:

$$\bar{x} = 2.16 \text{ m}, \quad \bar{y} = 0.80 \text{ m}, \quad s_X^2 = 1.533 \text{ m}^2, \quad s_Y^2 = 0.362 \text{ m}^2, \quad \rho = 0.90$$

(a) Using the theoretical conditional expectation for jointly normal variables $E(Y|X = x) = \mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X)$, express the slope β and intercept α in terms of ρ, s_X, s_Y, $\bar{x}$, and $\bar{y}$.

(b) Compute the linear regression line $\hat{y} = \hat{\alpha} + \hat{\beta}x$.

(c) Calculate the conditional variance $\text{Var}(Y|X = x) = s_Y^2(1 - \rho^2)$ and compare it directly to $s_{Y|X}^2 = \frac{\Delta^2}{n-2}$. ■